# Miguel Angel Torres-Montoya

Crime Economics · Education Economics · Applied Econometrics · Causal Inference
m.torresmontoya22@gmail.com | miguel.torres2@u.icesi.edu.co | +57 (316) 244-0219
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## EDUCATION

**Universidad Icesi,** Colombia.
Bachelor of Arts in Economics. GPA: 4.5/5.0. Expected 2027

**Graduate-level Coursework:**
Quantitative and Computational Methods (4.8/5.0),
Advanced Macroeconomics (4.8/5.0),
Advanced Microeconomics (in progress).
Game Theory and Asymmetric Information (in progress).
Advanced Econometrics (in progress).

**Universidad del Rosario,** Bogotá, Colombia.
Real Analysis, Bogotá Summer School in Economics (4.7/5.0). 2026

## PUBLISHED WORK

**Illicit Drugs in Colombia: The Limits of Eradication and the Tasks Ahead.**

With Juan David Gélvez and Eduard F. Martínez. Centro de Estudios sobre Seguridad y Drogas (CESED), Universidad de los Andes. Adapted in English as "Colombia's New, Old Drug Policy: Reviving a Failed Model," GIGA Focus Latin America, No. 7, 2026.

Examines the cocaine value chain from coca leaf to the European retail market, showing why eradication targets the lowest-value, most easily replaceable link. [Policy Brief (Spanish)] [GIGA Focus (English adaptation)]

**Quarterly Monitoring of Economic Activity in Bogotá with Satellite and Mobility Data.**

With Eduard F. Martínez-González.

Cuadernos de Desarrollo Económico No. 83, Secretaría de Desarrollo Económico, Alcaldía Mayor de Bogotá. February 2026

Quarterly GDP estimation at 450m grid level using VIIRS night lights, TransMilenio mobility, and census data; feedforward neural network trained on ~16,000 model configurations with rolling-origin validation. [Report]

## RESEARCH IN PROGRESS

**The Gender of Justice: Judges and Sentencing Disparities in Colombia.**

With Eduard F. Martínez and Daniel Mejía.

Undergraduate thesis. Causally identifies the effect of judge sex on sentence length using institutional random assignment (SARJ) across 24 Colombian judicial districts.

**State Capacity and Education Policy Implementation in Colombia.**

Staggered difference-in-differences design; examines whether local institutional capacity shapes the effect of the full-school-day reform on dropout

**Impact Evaluation of a Household-Based Poverty Reduction Program in Southwestern Colombia.**

With Eduard F. Martínez-González and Blanca Zuluaga. CIENFI, Universidad Icesi (consulting project for Fundación Carvajal).

Fuzzy regression discontinuity design combining administrative records with a new endline survey (fall 2026). Role: day-to-day project management, data processing, estimation, and results.

## Research Experience

**2025 – present** Research Assistant, Center for Research in Economics and Finance (CIENFI), Universidad Icesi, Cali, Colombia. Supervisor: Eduard F. Martínez-González.

Projects include quarterly GDP monitoring for Bogotá (Cuadernos de Desarrollo Económico No. 83), research on criminal sentencing and education policy, and the analysis below.

**2026 – present** Research Seedbed, Finance & Investment Club (FIC), Universidad Icesi, Cali, Colombia.

Contributing researcher in the club's newly founded research division. Designed and delivered a workshop on causal inference methods for undergraduate researchers. [Workshop Slides]

## Teaching Experience

**2025 – present** Teaching Monitor, Universidad Icesi, Cali, Colombia.

Introduction to Business Analytics (2025); Microeconomic Theory III, International Economics, Macroeconomic Theory I (2026).

Big Data and Machine Learning (2026) — graduate elective, M.A. in Economics.

## Awards and Honors

**2026** 1st Place, Undergraduate Presentation Competition on the Behavioral Economics, Universidad Icesi. [Slides]

**2026** Honor Scholarship, Universidad Icesi. Awarded for ranking first in GPA among Economics students (2026-1).

**2025** 3rd Place, Undergraduate Presentation Competition on the Economics of Crime, Universidad Icesi. [Slides]

**2023** 1st Place, Undergraduate Economic Debate Competition, Universidad Icesi.

**2022–2026** Honor Roll, Universidad Icesi. Recognized for academic excellence in every semester.

## Additional Training

**World Bank Group,** via edX.

IEM01x: Evaluating Impact in Low- and Middle-Income Countries. [Certificate] 2026

## TECHNICAL SKILLS

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| **Programming:** | R (Advanced), Python (Intermediate). |
| **Geospatial:** | Google Earth Engine (rgee), QGIS. |
| **Other Tools:** | Excel (Advanced), LaTeX, Git, PowerPoint, Canva. |
| **Methods:** | Machine Learning (XGBoost, k-NN, Neural Networks), Spatial Econometrics, Remote Sensing, Statistical Modeling, Data Visualization, Causal Inference. |

## LANGUAGES

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| **Spanish:** | Native. |
| **English:** | Professional Working Proficiency (C1). |